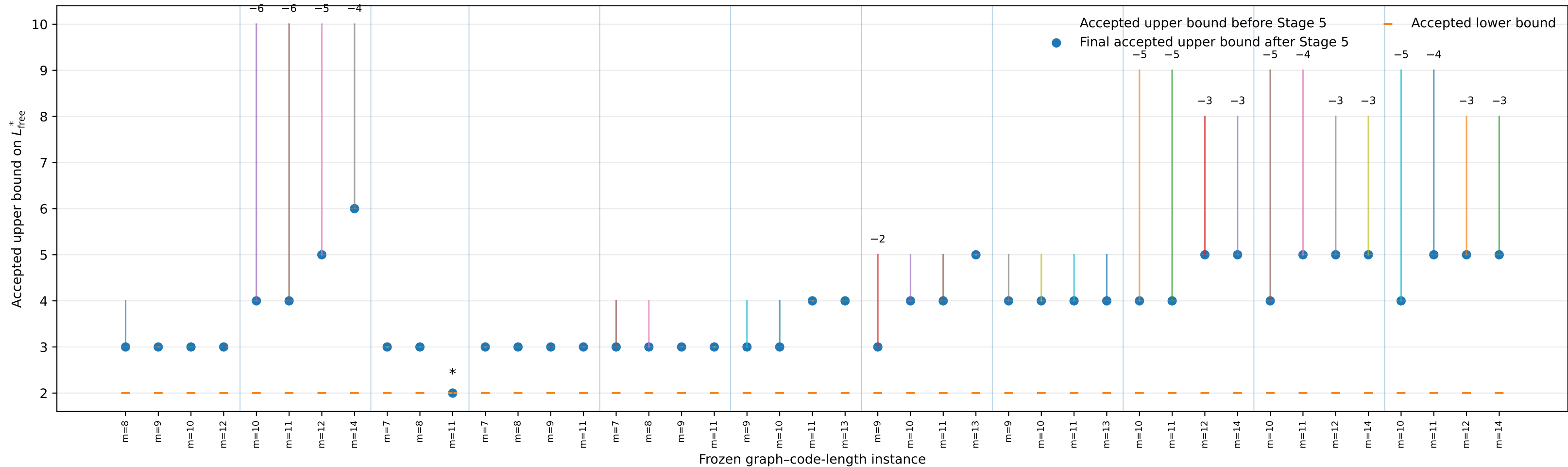


Stage 5 scalable free-codebook search

Accepted upper bounds before and after the frozen heuristic search

28 of 42 searched instances improved; one additional instance is exact by accepted Stage 4 transfer. Independent reproduction: 478/478 runs PASS.



* exact by accepted transfer. Heuristic witnesses establish feasible upper bounds; they do not by themselves establish optimality.